

HYDRAULIC TURBINES

SELECTION AND CLASSIFICATION OF TURBINES

- The type of the turbine can be determined if the head available, power to be developed and speed at which it has to run are known.
- The turbine is also selected based on head of water available at the turbine inlet.

■ Classification of hydraulic turbines

1. According to head and quantity of water available

- a) High head turbine – Head is more than 250 m. Eg: Pelton turbine
- b) Medium head turbine – 60 m to 250 m. Eg: Francis turbine
- c) Low head turbine – Head less than 60 m. Eg: Kaplan turbine

2. According to the direction of flow of water in the runner

- a) Tangential flow – Water strikes the wheel tangentially
- b) Radial flow – Water enters and leaves the turbine radially
- c) Mixed flow – Water enters radially but leaves axially
- d) Axial flow – Water flows parallel to the axis of the turbine shaft

3. According to action of water on moving blades

- a) Impulse turbine – Water possesses only kinetic energy at the inlet of the turbine and the turbine works at atmospheric pressure.
- b) Reaction turbine – water possesses both kinetic energy and pressure energy at the inlet.

4. According to specific speed

- a) Low specific speed – (eg. Pelton wheel, $N_s < 30$ for single jet, $30 < N_s < 51$ for multi jet)
- b) Medium specific speed – (eg. Francis turbine, N_s - 51 to 225)
- c) High specific speed – (eg. Kaplan turbine, N_s - 255 to 860)

- Hydraulic turbine is a device which converts the energy of water into mechanical energy.
 - It consists of a wheel called runner provided with a number of curved or straight vanes on its periphery.
 - Hydraulic turbines are classified on the basis of the action of water on these moving blades.
- ✓ Impulse turbine
 - ✓ Reaction turbine

IMPULSE TURBINE

- Potential energy of water is converted into kinetic energy by a set of nozzles.
- This produces powerful jets impinging on the vanes provided on the periphery of wheel.
- Examples of impulse turbine are Pelton turbine, Turgo wheel, Jonval turbine etc.
- This type of turbine is used where **high head of water** is available.
- Pelton turbine is the most commonly used impulse turbine.

PELTON TURBINE

- It is a tangential flow impulse turbine
- It is preferred when available water source has relatively high head at low flow rates

Construction:

1. Nozzle and flow regulating arrangement

- Water is transferred from high head source through penstock which is fitted with a nozzle. Through the nozzle, the water flows out as a high speed jet.
- A needle spear moving inside the nozzle controls the flow of water
- All the available potential energy of water is thus converted into kinetic energy

2. Runner and buckets

- Runner consists of a circular disc on the periphery of which a number of buckets are evenly spaced.
- The shape of the bucket is of double hemispherical cup.
- Each bucket is divided into two symmetrical parts by dividing wall known as splitter which divides the jet into two equal parts.



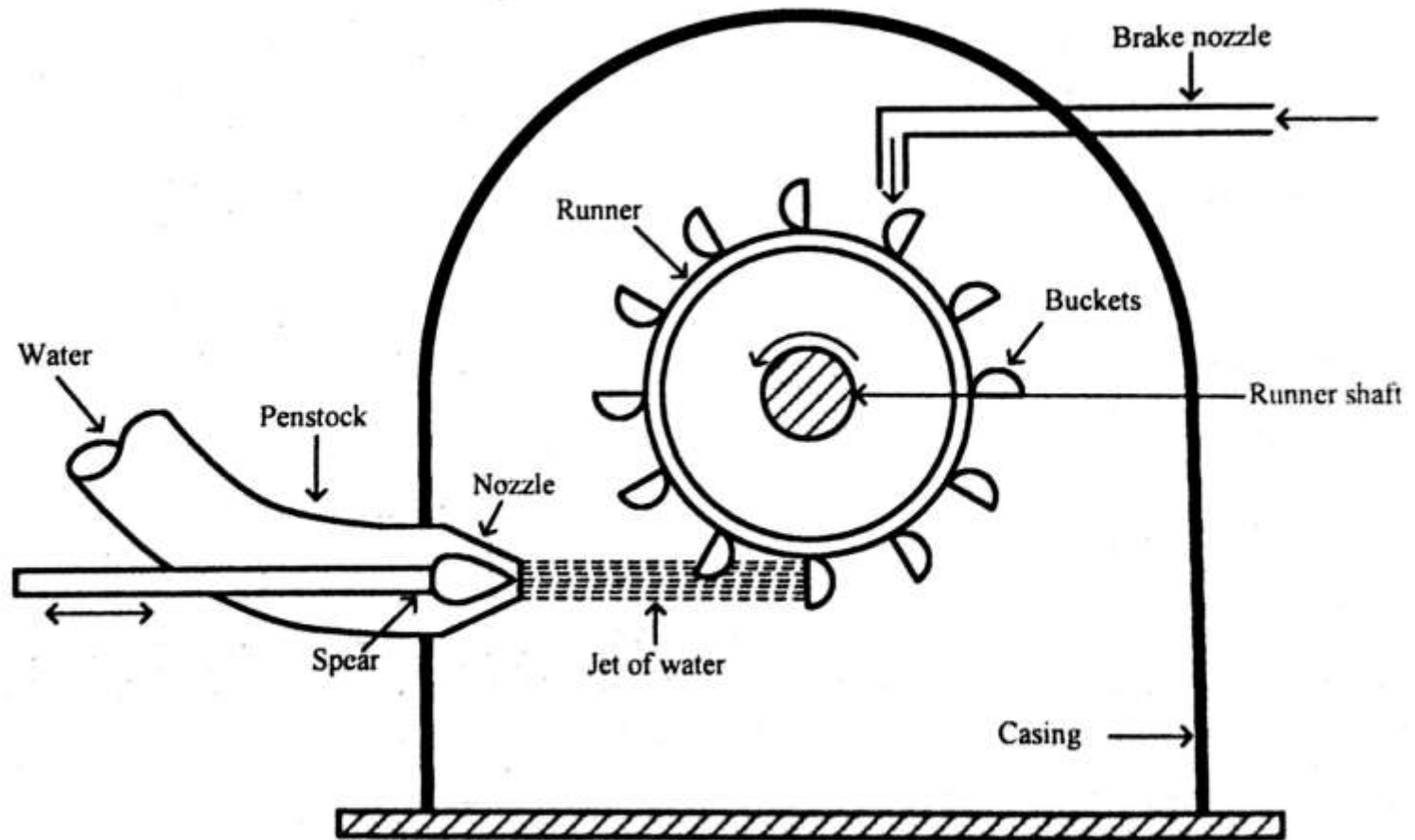
3. Casing

- To prevent the splashing of water and to discharge water to tailrace.

4. Breaking jet

- To stop the runner in a short time, a nozzle is provided which directs jet of water on the back of the vanes.

Pelton turbine



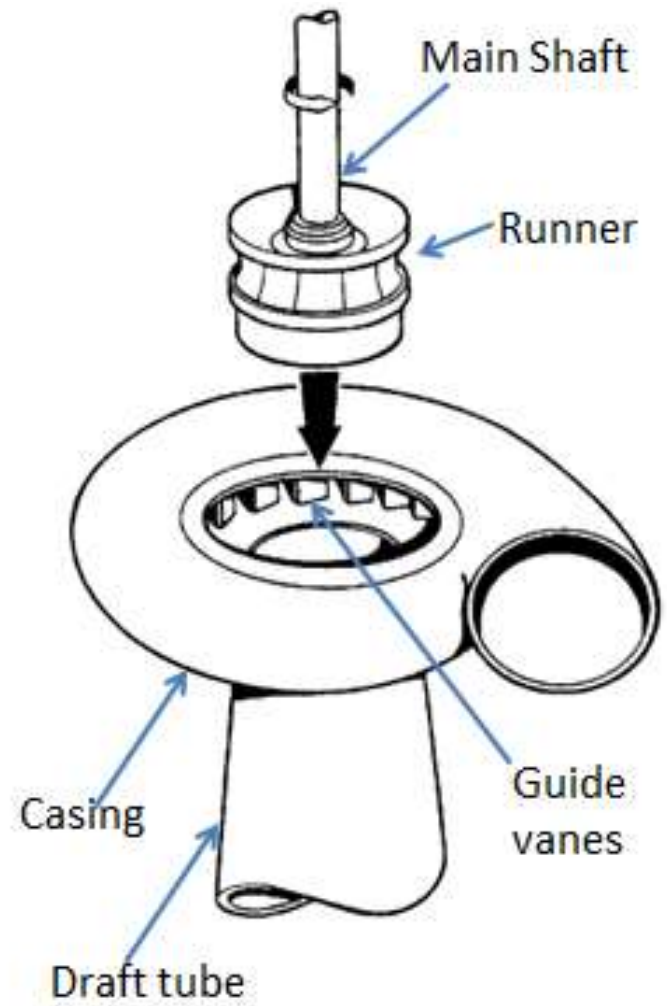
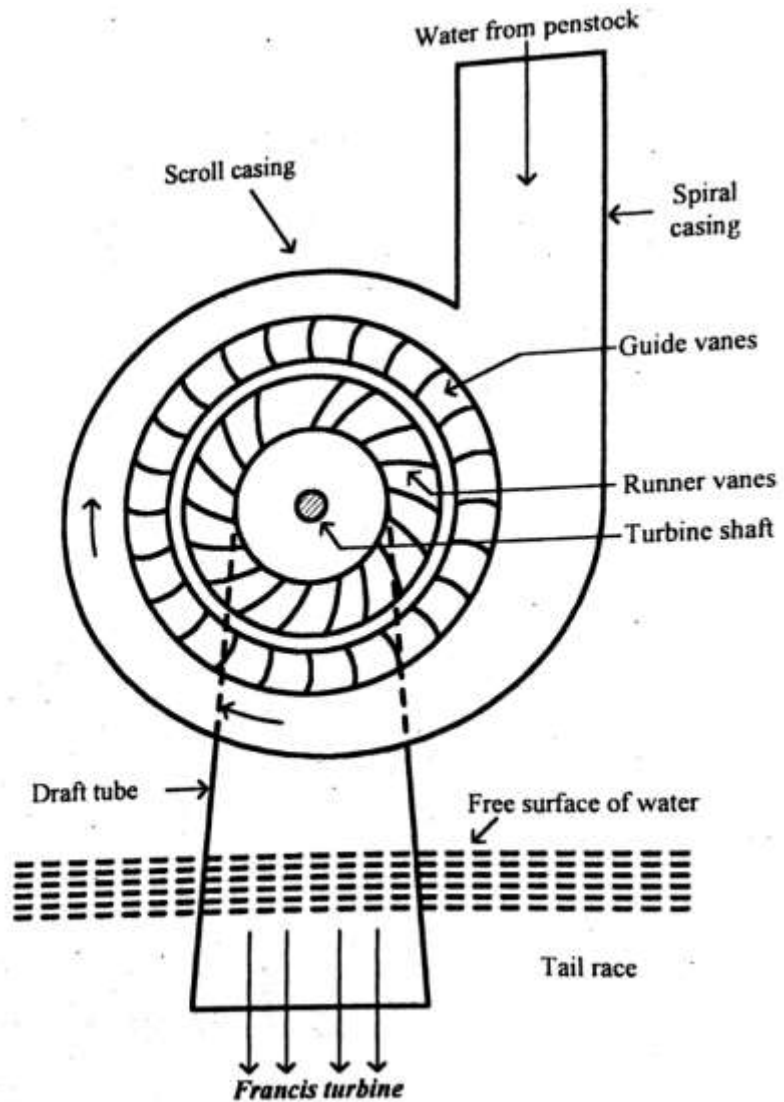
Cross sectional view of Pelton turbine

Working:

- Potential energy of water is converted into kinetic energy by a set of nozzles.
- This powerful jet impinges on the vanes provided on the periphery of runner wheel.
- The wheel is fixed to a shaft, which is coupled with a generator.
- The water after imparting its energy is discharged into the tailrace.

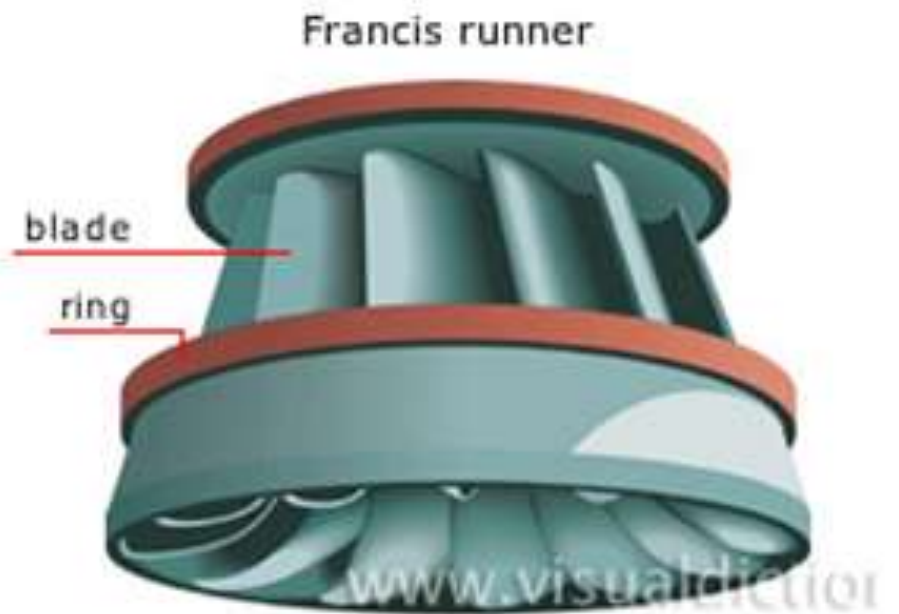
FRANCIS TURBINE

- Francis turbine is a mixed flow reaction turbine
- Water under pressure enters the runner from the guide vanes towards the centre in radial direction and discharges out of the runner axially.
- It operates under medium head (60 m – 250 m) and medium flow rate
- The head acting on the turbine is partly transformed to KE and rest remains as pressure head.



➤ CONSTRUCTION

1. **Spiral Casing** – Casing around the turbine runner, spiral in shape, and it evenly distributes the water around the circumference of the wheel
2. **Guide vanes** – These are fixed around the circumference of the runner. Each guide vane can rotate about its pivot centre and thus flow cross sectional area can be varied. This helps in regulating the quantity of water going to the runner.
3. **Runner and runner vanes** – Runner is a circular wheel in which a series of radial curved vanes are fixed.
4. **Draft tube** – The exit end of the runner is connected to the small end of draft tube

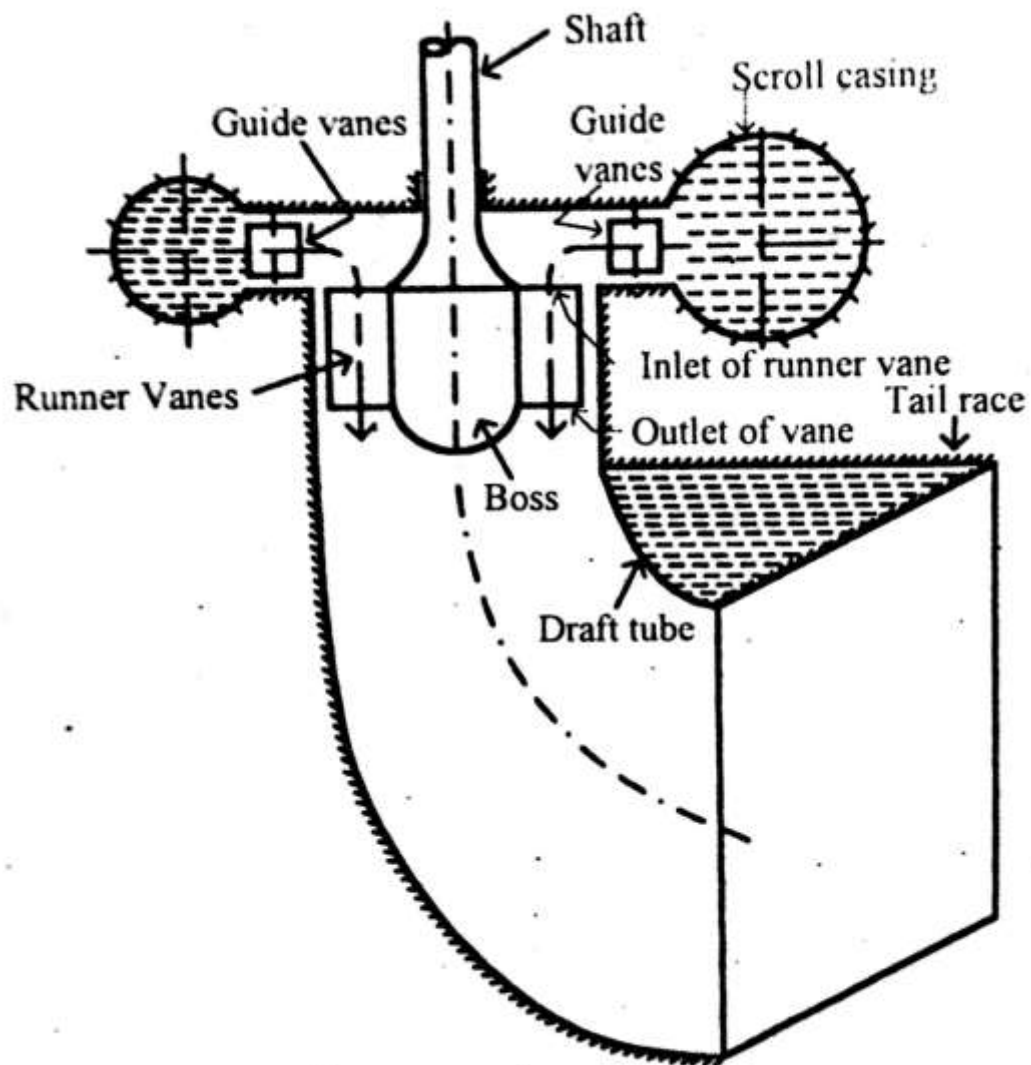


➤ Working

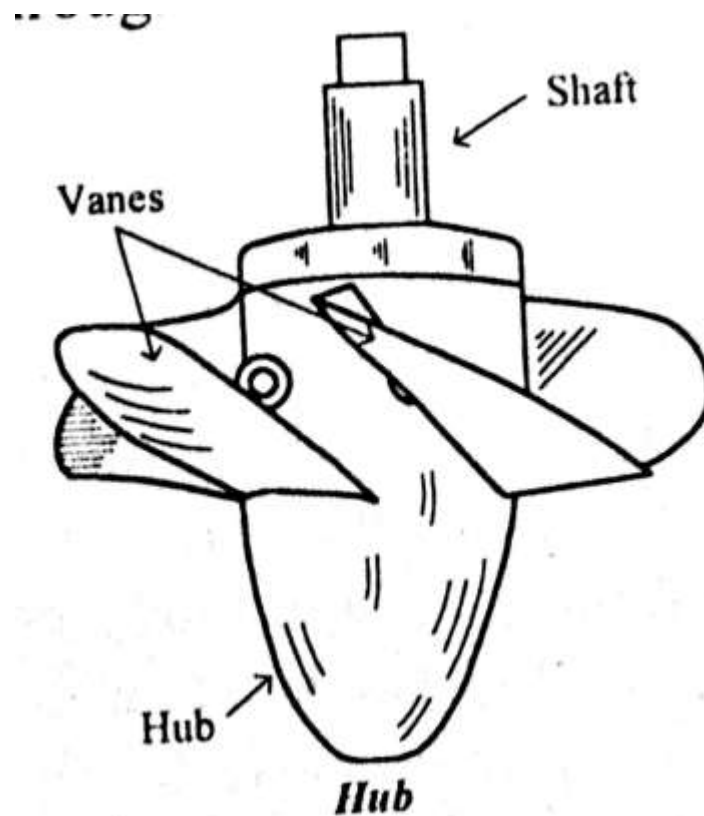
- Water from the penstock is brought down to the turbine at high pressure through spiral casing.
- The guide vanes have profiles like aerofoil so that the flow is smooth and they behave like nozzles.
- The water leaves the guide vanes in such a way that it enters the runner blades radially to the runner.
- As the water passes through the runner its pressure changes gradually, being higher at the inlet than at the outlet. The difference in pressure is known as reaction pressure and is responsible for the rotation of the runner.
- After doing work, the water is discharged to the tailrace axially from the centre of the turbine through a draft tube.

KAPLAN TURBINE

- Kaplan turbine is widely used in high flow- low head power generation.
- For a runner of given diameter, the maximum flow rate is achieved when the flow is parallel to the axis. Such a machine is known as axial flow reaction turbine.
- They are very expensive to design, manufacture and install, but operates for decades.
- It is a reaction turbine in which the water at the inlet flows through runner along the direction parallel to the axis of rotation of runner



Main parts of Kaplan turbine



➤ CONSTRUCTION

- Scroll casing – Water from the penstock enters the scroll casing and then moves to the guide vanes. From the guide vanes, water turns through 90 degrees and flows axially through runner.
- Guide vane mechanism
- Hub – The lower end of the shaft is made longer and is called hub or boss. The vanes are fixed on the hub and hence acts as runner. These blades can be adjusted according to the direction of flow and available water head. The diameter of runner is typically 5m to 8m.
- Draft tube

DRAFT TUBE

- Draft tube is a passage which connects the runner exit to the tail race .
- Draft tube is made divergent so as to reduce the velocity at the outlet of draft tube to minimum.
- The primary function of the draft tube is to reduce the velocity of discharged water to minimise loss of kinetic energy at the outlet.

➤ The functions of draft tube are:

- It makes the installation possible above the tail race level without the loss of head.
- Loss of kinetic energy is converted into pressure energy
- It prevents the splashing of water coming out of the runner.

Comparison of impulse and reaction turbine

No.	Aspects	Impulse turbine	Reaction turbine
1	Conversion of fluid energy	The available energy is converted into KE by nozzles	The energy of fluid is partly converted into KE and the rest remains as pressure head
2	Availability of head and discharge	Requires high head and small quantity of flow	Requires low head and high quantity of flow
3	Changes in pressure	Pressure remains same throughout the action of water	While passing through the runner water undergoes changes in velocity and pressure
4	Draft tube	Not required	Pressure of water in the turbine which is lower than the atmospheric at exit is increased by a draft tube
5	Admittance of water over the wheel	Water jet strikes a few buckets at a time with kinetic energy	Water is guided by the guide vanes to flow over the blades with pressure energy
6	Example	Pelton turbine	Francis turbine, Kaplan turbine